

Day 1 Learning Summary: Database Fundamentals

Introduction

A **database** is an organized collection of data that allows information to be stored, accessed, managed, and updated efficiently. Databases are the foundation of modern applications such as banking systems, e-commerce platforms, social media, healthcare, and business applications.

1. Database, DBMS, and RDBMS

Term	Description
Database	A structured collection of related data.
DBMS (Database Management System)	Software used to create, store, retrieve, update, and manage databases.
RDBMS (Relational Database Management System)	A DBMS that stores data in related tables using rows and columns and supports SQL.

Popular RDBMS

- MySQL
 - PostgreSQL
 - Microsoft SQL Server
 - Oracle Database
 - SQLite
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2. Why Databases Matter for Data Analysts

Databases are essential because:

- Business data is stored in databases rather than spreadsheets.
 - They can manage millions or billions of records efficiently.
 - SQL enables fast and accurate data retrieval.
 - Data extraction is the first step in most analytics and reporting workflows.
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3. Relational vs Non-Relational Databases

Relational (SQL)	Non-Relational (NoSQL)
Data stored in tables	Data stored as documents, key-value pairs, graphs, etc.
Fixed schema	Flexible schema
Uses SQL	Uses NoSQL technologies
Best for structured business data	Best for rapidly changing or unstructured data

Examples

Relational

- MySQL
- PostgreSQL
- SQL Server
- Oracle
- SQLite

Non-Relational

- MongoDB
- Redis
- Cassandra
- DynamoDB

4. Database Terminology

Table

A collection of related records organized into rows and columns.

Example:

Customer_ID	Name	City
101	John	Toronto

Column (Field/Attribute)

Represents one type of information.

Examples:

- Customer_ID
 - Name
 - City
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Row (Record/Tuple)

A complete record in a table.

Example:

| 101 | John | Toronto |

Schema

The blueprint of a database containing:

- Tables
 - Columns
 - Data types
 - Relationships
 - Constraints
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Primary Key (PK)

A column that uniquely identifies every row.

Characteristics:

- Unique
- Cannot contain NULL values
- One primary key per table

Example:

Customer_ID (PK)	Name
101	John

Foreign Key (FK)

A column that references the primary key of another table.

Example:

Customers

Customer_ID	Name
101	John

Orders

Order_ID	Customer_ID
5001	101

Here, **Customer_ID** in the Orders table is the Foreign Key.

Relationship

A logical connection between two tables established using Primary Keys and Foreign Keys.

Example:

Customers → Orders

One customer can place multiple orders.

5. Introduction to Normalization

Definition

Normalization is the process of organizing data to:

- Reduce redundancy

- Improve consistency
 - Eliminate duplicate data
 - Maintain data integrity
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Before Normalization

Order_ID	Customer_Name	City	Product
101	John	Toronto	Laptop
102	John	Toronto	Mouse

Problems:

- Duplicate customer information
 - Difficult to update records
 - Higher storage requirements
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After Normalization

Customers

Customer_ID	Name	City
1	John	Toronto

Orders

Order_ID	Customer_ID	Product
101	1	Laptop
102	1	Mouse

Benefits:

- No duplicated customer data
 - Easier updates
 - Better consistency
 - Efficient storage
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6. Key Takeaways

- A **Database** stores organized data.

- A **DBMS** manages databases.
 - An **RDBMS** stores data in related tables using SQL.
 - Data analysts use SQL to retrieve and analyze business data.
 - Relational databases organize information using rows and columns.
 - **Primary Keys** uniquely identify records.
 - **Foreign Keys** connect related tables.
 - **Relationships** allow data to be combined across multiple tables.
 - **Normalization** reduces redundancy and improves data quality.
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Day 1 Quick Recap

- ✓ What is a Database?
 - ✓ Difference between Database, DBMS, and RDBMS
 - ✓ Importance of Databases in Data Analytics
 - ✓ Relational vs NoSQL Databases
 - ✓ Table, Column, Row, and Schema
 - ✓ Primary Key and Foreign Key
 - ✓ Relationships between Tables
 - ✓ Introduction to Normalization
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Interview Questions

1. What is a database?
2. What is the difference between DBMS and RDBMS?
3. Why are databases important for data analysts?
4. What is the difference between SQL and NoSQL databases?
5. What is a table?
6. What is a schema?
7. What is a Primary Key?
8. What is a Foreign Key?
9. What is normalization?
10. Why is normalization important?